# Martim Vicente

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## Education

**Instituto Superior Técnico** – BSc in Computer Science — Sep. 2024 – Jun. 2027
**Average:** 16.00 / 20.00
**Awards:** Gulbenkian Merit Scholarship, Técnico ALUMNI Scholarship, WARD Analytics Scholarship

## Experience

**Software Team Lead,** Rocket Experience Division – Lisbon, Portugal — May 2026 – Present

- Re-architected a legacy STM32 Ignition Software into a modular FreeRTOS-based platform, replacing an unmaintainable codebase and enabling scalable development.
- Mentored and co-developed the Ignition Software a junior engineer during the redesign, establishing software architecture, coding standards, and development workflows.

**Software Engineer,** Rocket Experience Division – Lisbon, Portugal — Sep. 2025 – May 2026

- Architected a STM32-based remote fueling platform using Hexagonal Architecture, isolating business logic from hardware drivers.
- Developed hardware abstraction layers in C for sensors, valves, and communication peripherals, enabling simulation and hardware-independent testing.
- Built a centralized MkDocs documentation platform deployed through GitHub Actions and Cloudflare Pages, automating documentation generation from code repositories and KiCad projects.

**Software Engineer,** Técnico Investment Club – Lisbon, Portugal — Sep. 2025 – May 2026

- Architected a low-latency Trading Engine in C++ execution and decoupled strategy, market data, and order execution layers.
- Replaced a legacy Python Trading Engine with a decoupled ETL and Execution architecture, containerizing each strategy into its own Docker pod to achieve fault isolation and prevent global system crashes.

## Projects

**Plume** — github.com/martimvct/plume

- Developed a logging library focused on to record high-rate sensor telemetry and flight events in real-time, eliminating timing delays that could otherwise destabilize active flight-control loops.
- Prevented mid-flight memory crashes and data loss by implementing a statically allocated queue (zero-heap) with a non-blocking overflow policy, ensuring deterministic RAM utilization during high-frequency diagnostic bursts.

**Mivros** — github.com/martimvct/mivros

- Architected a local-first inventory management system using Flutter and SQLite to solve manual tracking bottlenecks, implementing a custom finite state machine that guarantees strict atomic transitions across the product lifecycle.
- Reduced manual data entry friction by integrating on-device ISBN barcode scanning and clipboard parsing, decreasing average logging time per item and ensuring highly accurate revenue and margin calculations.

## Skills

**Languages:** C, C++, Python, Go, Java, RISC-V Assembly
**Platforms:** STM32, ESP32, Raspberry Pi
**Tools:** CMake, STM32Cube, Docker, Github Actions, Git